DSC550Week3

June 28, 2025

0.0.1 Loading required libraries for textblob and vaderSentiment

[189]:
```
pip install textblob
```

```
Requirement already satisfied: textblob in /opt/anaconda3/lib/python3.12/site-
packages (0.19.0)
Requirement already satisfied: nltk>=3.9 in /opt/anaconda3/lib/python3.12/site-
packages (from textblob) (3.9.1)
Requirement already satisfied: click in /opt/anaconda3/lib/python3.12/site-
packages (from nltk>=3.9->textblob) (8.1.7)
Requirement already satisfied: joblib in /opt/anaconda3/lib/python3.12/site-
packages (from nltk>=3.9->textblob) (1.4.2)
Requirement already satisfied: regex>=2021.8.3 in
/opt/anaconda3/lib/python3.12/site-packages (from nltk>=3.9->textblob)
(2023.10.3)
Requirement already satisfied: tqdm in /opt/anaconda3/lib/python3.12/site-
packages (from nltk>=3.9->textblob) (4.66.4)
Note: you may need to restart the kernel to use updated packages.
```

[191]:
```
pip install vaderSentiment
```

```
Requirement already satisfied: vaderSentiment in
/opt/anaconda3/lib/python3.12/site-packages (3.3.2)
Requirement already satisfied: requests in /opt/anaconda3/lib/python3.12/site-
packages (from vaderSentiment) (2.32.2)
Requirement already satisfied: charset-normalizer<4,>=2 in
/opt/anaconda3/lib/python3.12/site-packages (from requests->vaderSentiment)
(2.0.4)
Requirement already satisfied: idna<4,>=2.5 in
/opt/anaconda3/lib/python3.12/site-packages (from requests->vaderSentiment)
(3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/opt/anaconda3/lib/python3.12/site-packages (from requests->vaderSentiment)
(2.2.2)
Requirement already satisfied: certifi>=2017.4.17 in
/opt/anaconda3/lib/python3.12/site-packages (from requests->vaderSentiment)
(2025.4.26)
Note: you may need to restart the kernel to use updated packages.
```

0.0.2 Loading packages required for this week

[193]:
```
import numpy as np
#import tensorflow as tf
#from tensorflow import keras
import pandas as pd
import seaborn as sns
from pylab import rcParams
import string
import re
import matplotlib.pyplot as plt
import math
from matplotlib import rc
from matplotlib.ticker import FixedLocator

from sklearn.model_selection import train_test_split
from collections import Counter, defaultdict
from bs4 import BeautifulSoup
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report, confusion_matrix
from nltk.stem.porter import PorterStemmer
from nltk.stem import WordNetLemmatizer
import nltk
from nltk.corpus import stopwords
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from IPython.display import display

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.ensemble import RandomForestClassifier
nltk.download('vader_lexicon')
sid = SentimentIntensityAnalyzer()


%matplotlib inline

sns.set(style='whitegrid', palette='muted', font_scale=1.5)

rcParams['figure.figsize'] = 14, 8

RANDOM_SEED = 42

np.random.seed(RANDOM_SEED)
nltk.download('stopwords')
```

```
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     /Users/vijsharm/nltk_data…
[nltk_data]   Package vader_lexicon is already up-to-date!
[nltk_data] Downloading package stopwords to
```

```
[nltk_data]     /Users/vijsharm/nltk_data…
[nltk_data]   Package stopwords is already up-to-date!
```

[193]: True

0.0.3 Loading required source file "labeledTrainData" for analysis

[198]:
```python
dftrain = pd.read_csv("labeledTrainData.tsv", header=0, \
                      delimiter="\t", quoting=3)
dftrain.shape
```

[198]: (25000, 3)

[200]:
```python
dftrain.head()
```

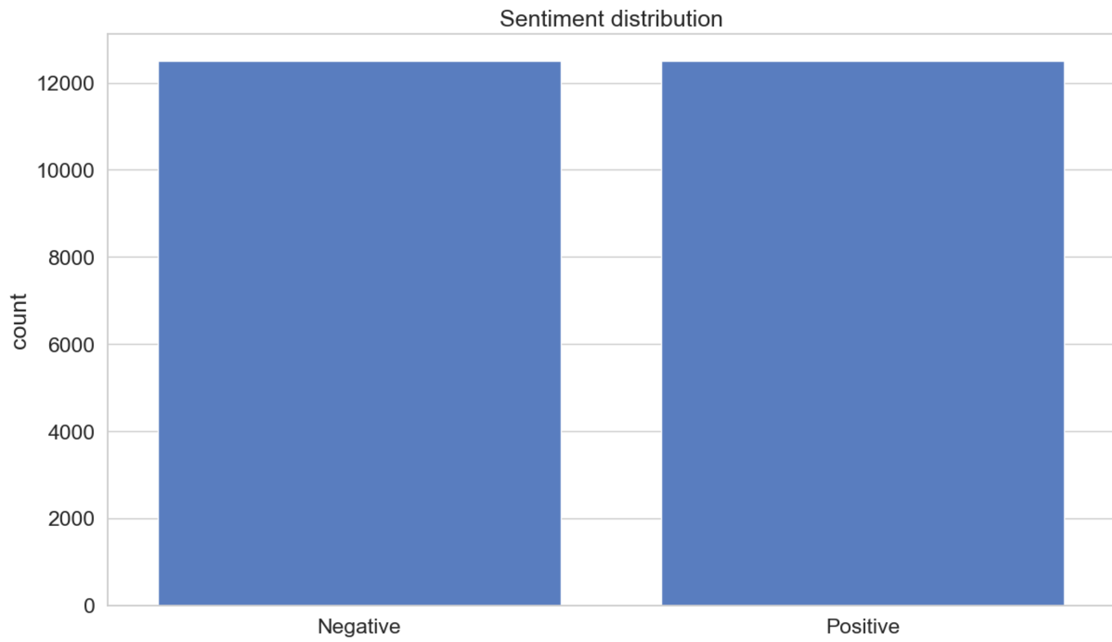
[200]:

	id	sentiment	review
0	"5814_8"	1	"With all this stuff going down at the moment …
1	"2381_9"	1	"\"The Classic War of the Worlds\" by Timothy …
2	"7759_3"	0	"The film starts with a manager (Nicholas Bell…
3	"3630_4"	0	"It must be assumed that those who praised thi…
4	"9495_8"	1	"Superbly trashy and wondrously unpretentious …

0.0.4 How many of each positive and negative reviews are there?

[203]:
```python
f = sns.countplot(x='sentiment', data=dftrain)
f.set_title("Sentiment distribution")
f.set_xticklabels(['Negative', 'Positive'])
plt.xlabel("");
```

```
/var/folders/18/ygzmjvj90m72g8_1d1sfk1b80000gn/T/ipykernel_1364/961267401.py:3:
UserWarning: set_ticklabels() should only be used with a fixed number of ticks,
i.e. after set_ticks() or using a FixedLocator.
  f.set_xticklabels(['Negative', 'Positive'])
```



0.0.5 Use TextBlob to classify each movie review as positive or negative. Assume that a polarity score greater than or equal to zero is a positive sentiment and less than 0 is a negative sentiment.

[206]:
```python
from textblob import TextBlob
import pandas as pd

def classify_sentiment(review):
    """
    Classifies a movie review as positive or negative based on its TextBlob␣
 ↪polarity score.

    Args:
        review (str): The movie review text.

    Returns:
        str: "Positive" if the polarity score is greater than or equal to 0,␣
 ↪"Negative" otherwise.
    """
    analysis = TextBlob(review)
    if analysis.sentiment.polarity >= 0:
        return "Positive"
    else:
        return "Negative"
```

```
# Apply the sentiment classification function to each review
dftrain["predicted_sentiment"] = dftrain["review"].apply(classify_sentiment)

# Evaluate the model (replace with your desired evaluation method)
# For example, comparing predicted_sentiment with the actual 'sentiment' column
accuracy = (dftrain["predicted_sentiment"] == dftrain["sentiment"]).mean()

print(f"Accuracy: {accuracy}")

# You can also print a few examples:
print("\nExample classifications:")
print(dftrain[["review", "predicted_sentiment"]].head())
```

```
Accuracy: 0.0

Example classifications:
                                              review predicted_sentiment
0  "With all this stuff going down at the moment …            Positive
1  "\"The Classic War of the Worlds\" by Timothy …            Positive
2  "The film starts with a manager (Nicholas Bell…            Negative
3  "It must be assumed that those who praised thi…            Positive
4  "Superbly trashy and wondrously unpretentious …            Negative
```

0.0.6 For up to five points extra credit, use another prebuilt text sentiment analyzer, e.g., VADER, and repeat steps (3) and (4).

[208]:
```
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
```

[211]:
```
def analyze_sentiment_vader(sentence):
    analyzer = SentimentIntensityAnalyzer()
    scores = analyzer.polarity_scores(sentence)
    compound_score = scores['compound']
    if compound_score >= 0.05:
        return 'positive'
    elif compound_score <= -0.05:
        return 'negative'
    else:
        return 'neutral'
```

[213]:
```
dftrain['vader_sentiment'] = dftrain['review'].apply(analyze_sentiment_vader)
vader_accuracy = (dftrain['sentiment'] == dftrain['vader_sentiment']).mean()
print(f'VADER Accuracy: {vader_accuracy}')
```

```
VADER Accuracy: 0.0
```

[214]:
```
print(dftrain.review[0])
```

```
"With all this stuff going down at the moment with MJ i've started listening to
his music, watching the odd documentary here and there, watched The Wiz and
```

```
watched Moonwalker again. Maybe i just want to get a certain insight into this
guy who i thought was really cool in the eighties just to maybe make up my mind
whether he is guilty or innocent. Moonwalker is part biography, part feature
film which i remember going to see at the cinema when it was originally
released. Some of it has subtle messages about MJ's feeling towards the press
and also the obvious message of drugs are bad m'kay.<br /><br />Visually
impressive but of course this is all about Michael Jackson so unless you
remotely like MJ in anyway then you are going to hate this and find it boring.
Some may call MJ an egotist for consenting to the making of this movie BUT MJ
and most of his fans would say that he made it for the fans which if true is
really nice of him.<br /><br />The actual feature film bit when it finally
starts is only on for 20 minutes or so excluding the Smooth Criminal sequence
and Joe Pesci is convincing as a psychopathic all powerful drug lord. Why he
wants MJ dead so bad is beyond me. Because MJ overheard his plans? Nah, Joe
Pesci's character ranted that he wanted people to know it is he who is supplying
drugs etc so i dunno, maybe he just hates MJ's music.<br /><br />Lots of cool
things in this like MJ turning into a car and a robot and the whole Speed Demon
sequence. Also, the director must have had the patience of a saint when it came
to filming the kiddy Bad sequence as usually directors hate working with one kid
let alone a whole bunch of them performing a complex dance scene.<br /><br
/>Bottom line, this movie is for people who like MJ on one level or another
(which i think is most people). If not, then stay away. It does try and give off
a wholesome message and ironically MJ's bestest buddy in this movie is a girl!
Michael Jackson is truly one of the most talented people ever to grace this
planet but is he guilty? Well, with all the attention i've gave this
subject…hmmm well i don't know because people can be different behind closed
doors, i know this for a fact. He is either an extremely nice but stupid guy or
one of the most sickest liars. I hope he is not the latter."
```

0.0.7 Removing HTML Markup with The BeautifulSoup Package

Create BeautifulSoup object to parse HTML.

[220]:
```python
dftrain['review_bs'] = dftrain['review'].apply(lambda x: BeautifulSoup(x, 'html.
↪parser'))
```

```
/var/folders/18/ygzmjvj90m72g8_1d1sfk1b80000gn/T/ipykernel_1364/1217238460.py:1:
MarkupResemblesLocatorWarning: The input looks more like a filename than markup.
You may want to open this file and pass the filehandle into Beautiful Soup.
  dftrain['review_bs'] = dftrain['review'].apply(lambda x: BeautifulSoup(x,
'html.parser'))
```

[221]:
```python
dftrain.review_bs[0].get_text()
```

[221]:
```
'"With all this stuff going down at the moment with MJ i\'ve started listening
to his music, watching the odd documentary here and there, watched The Wiz and
watched Moonwalker again. Maybe i just want to get a certain insight into this
guy who i thought was really cool in the eighties just to maybe make up my mind
whether he is guilty or innocent. Moonwalker is part biography, part feature
```

```
film which i remember going to see at the cinema when it was originally
released. Some of it has subtle messages about MJ\'s feeling towards the press
and also the obvious message of drugs are bad m\'kay.Visually impressive but of
course this is all about Michael Jackson so unless you remotely like MJ in
anyway then you are going to hate this and find it boring. Some may call MJ an
egotist for consenting to the making of this movie BUT MJ and most of his fans
would say that he made it for the fans which if true is really nice of him.The
actual feature film bit when it finally starts is only on for 20 minutes or so
excluding the Smooth Criminal sequence and Joe Pesci is convincing as a
psychopathic all powerful drug lord. Why he wants MJ dead so bad is beyond me.
Because MJ overheard his plans? Nah, Joe Pesci\'s character ranted that he
wanted people to know it is he who is supplying drugs etc so i dunno, maybe he
just hates MJ\'s music.Lots of cool things in this like MJ turning into a car
and a robot and the whole Speed Demon sequence. Also, the director must have had
the patience of a saint when it came to filming the kiddy Bad sequence as
usually directors hate working with one kid let alone a whole bunch of them
performing a complex dance scene.Bottom line, this movie is for people who like
MJ on one level or another (which i think is most people). If not, then stay
away. It does try and give off a wholesome message and ironically MJ\'s bestest
buddy in this movie is a girl! Michael Jackson is truly one of the most talented
people ever to grace this planet but is he guilty? Well, with all the attention
i\'ve gave this subject…hmmm well i don\'t know because people can be
different behind closed doors, i know this for a fact. He is either an extremely
nice but stupid guy or one of the most sickest liars. I hope he is not the
latter."'
```

0.0.8 Dealing with Punctuation, Numbers and Stopwords: NLTK and regular expressions

[225]:
```python
dftrain['review_letters_only'] = dftrain['review_bs'].apply(lambda x: re.
 ↪sub(r'[^a-zA-Z]', ' ', x.get_text()))
```

[226]:
```python
dftrain['review_letters_only'][0]
```

[226]:
```
' With all this stuff going down at the moment with MJ i ve started listening to
his music  watching the odd documentary here and there  watched The Wiz and
watched Moonwalker again  Maybe i just want to get a certain insight into this
guy who i thought was really cool in the eighties just to maybe make up my mind
whether he is guilty or innocent  Moonwalker is part biography  part feature
film which i remember going to see at the cinema when it was originally released
Some of it has subtle messages about MJ s feeling towards the press and also the
obvious message of drugs are bad m kay Visually impressive but of course this is
all about Michael Jackson so unless you remotely like MJ in anyway then you are
going to hate this and find it boring  Some may call MJ an egotist for
consenting to the making of this movie BUT MJ and most of his fans would say
that he made it for the fans which if true is really nice of him The actual
feature film bit when it finally starts is only on for    minutes or so
```

```
excluding the Smooth Criminal sequence and Joe Pesci is convincing as a
psychopathic all powerful drug lord  Why he wants MJ dead so bad is beyond me
Because MJ overheard his plans  Nah  Joe Pesci s character ranted that he wanted
people to know it is he who is supplying drugs etc so i dunno  maybe he just
hates MJ s music Lots of cool things in this like MJ turning into a car and a
robot and the whole Speed Demon sequence  Also  the director must have had the
patience of a saint when it came to filming the kiddy Bad sequence as usually
directors hate working with one kid let alone a whole bunch of them performing a
complex dance scene Bottom line  this movie is for people who like MJ on one
level or another  which i think is most people   If not  then stay away  It does
try and give off a wholesome message and ironically MJ s bestest buddy in this
movie is a girl  Michael Jackson is truly one of the most talented people ever
to grace this planet but is he guilty  Well  with all the attention i ve gave
this subject    hmmm well i don t know because people can be different behind
closed doors  i know this for a fact  He is either an extremely nice but stupid
guy or one of the most sickest liars  I hope he is not the latter  '
```

[229]:
```
dftrain['review_words'] = dftrain['review_letters_only'].apply(lambda x: x.
↪lower().split())
```

[231]:
```
dftrain['review_words'][0]
```

[231]:
```
['with',
 'all',
 'this',
 'stuff',
 'going',
 'down',
 'at',
 'the',
 'moment',
 'with',
 'mj',
 'i',
 've',
 'started',
 'listening',
 'to',
 'his',
 'music',
 'watching',
 'the',
 'odd',
 'documentary',
 'here',
 'and',
 'there',
```

```
'watched',
'the',
'wiz',
'and',
'watched',
'moonwalker',
'again',
'maybe',
'i',
'just',
'want',
'to',
'get',
'a',
'certain',
'insight',
'into',
'this',
'guy',
'who',
'i',
'thought',
'was',
'really',
'cool',
'in',
'the',
'eighties',
'just',
'to',
'maybe',
'make',
'up',
'my',
'mind',
'whether',
'he',
'is',
'guilty',
'or',
'innocent',
'moonwalker',
'is',
'part',
'biography',
'part',
'feature',
```

```
'film',
'which',
'i',
'remember',
'going',
'to',
'see',
'at',
'the',
'cinema',
'when',
'it',
'was',
'originally',
'released',
'some',
'of',
'it',
'has',
'subtle',
'messages',
'about',
'mj',
's',
'feeling',
'towards',
'the',
'press',
'and',
'also',
'the',
'obvious',
'message',
'of',
'drugs',
'are',
'bad',
'm',
'kay',
'visually',
'impressive',
'but',
'of',
'course',
'this',
'is',
'all',
```

```
'about',
'michael',
'jackson',
'so',
'unless',
'you',
'remotely',
'like',
'mj',
'in',
'anyway',
'then',
'you',
'are',
'going',
'to',
'hate',
'this',
'and',
'find',
'it',
'boring',
'some',
'may',
'call',
'mj',
'an',
'egotist',
'for',
'consenting',
'to',
'the',
'making',
'of',
'this',
'movie',
'but',
'mj',
'and',
'most',
'of',
'his',
'fans',
'would',
'say',
'that',
'he',
```

```
'made',
'it',
'for',
'the',
'fans',
'which',
'if',
'true',
'is',
'really',
'nice',
'of',
'him',
'the',
'actual',
'feature',
'film',
'bit',
'when',
'it',
'finally',
'starts',
'is',
'only',
'on',
'for',
'minutes',
'or',
'so',
'excluding',
'the',
'smooth',
'criminal',
'sequence',
'and',
'joe',
'pesci',
'is',
'convincing',
'as',
'a',
'psychopathic',
'all',
'powerful',
'drug',
'lord',
'why',
```

```
'he',
'wants',
'mj',
'dead',
'so',
'bad',
'is',
'beyond',
'me',
'because',
'mj',
'overheard',
'his',
'plans',
'nah',
'joe',
'pesci',
's',
'character',
'ranted',
'that',
'he',
'wanted',
'people',
'to',
'know',
'it',
'is',
'he',
'who',
'is',
'supplying',
'drugs',
'etc',
'so',
'i',
'dunno',
'maybe',
'he',
'just',
'hates',
'mj',
's',
'music',
'lots',
'of',
'cool',
```

```
'things',
'in',
'this',
'like',
'mj',
'turning',
'into',
'a',
'car',
'and',
'a',
'robot',
'and',
'the',
'whole',
'speed',
'demon',
'sequence',
'also',
'the',
'director',
'must',
'have',
'had',
'the',
'patience',
'of',
'a',
'saint',
'when',
'it',
'came',
'to',
'filming',
'the',
'kiddy',
'bad',
'sequence',
'as',
'usually',
'directors',
'hate',
'working',
'with',
'one',
'kid',
'let',
```

```
'alone',
'a',
'whole',
'bunch',
'of',
'them',
'performing',
'a',
'complex',
'dance',
'scene',
'bottom',
'line',
'this',
'movie',
'is',
'for',
'people',
'who',
'like',
'mj',
'on',
'one',
'level',
'or',
'another',
'which',
'i',
'think',
'is',
'most',
'people',
'if',
'not',
'then',
'stay',
'away',
'it',
'does',
'try',
'and',
'give',
'off',
'a',
'wholesome',
'message',
'and',
```

```
'ironically',
'mj',
's',
'bestest',
'buddy',
'in',
'this',
'movie',
'is',
'a',
'girl',
'michael',
'jackson',
'is',
'truly',
'one',
'of',
'the',
'most',
'talented',
'people',
'ever',
'to',
'grace',
'this',
'planet',
'but',
'is',
'he',
'guilty',
'well',
'with',
'all',
'the',
'attention',
'i',
've',
'gave',
'this',
'subject',
'hmmm',
'well',
'i',
'don',
't',
'know',
'because',
```

```
 'people',
 'can',
 'be',
 'different',
 'behind',
 'closed',
 'doors',
 'i',
 'know',
 'this',
 'for',
 'a',
 'fact',
 'he',
 'is',
 'either',
 'an',
 'extremely',
 'nice',
 'but',
 'stupid',
 'guy',
 'or',
 'one',
 'of',
 'the',
 'most',
 'sickest',
 'liars',
 'i',
 'hope',
 'he',
 'is',
 'not',
 'the',
 'latter']
```

[233]:
```
set_of_stopwords = set(stopwords.words("english"))
dftrain['review_meaningful_words'] = dftrain['review_words'].apply(lambda x: [w␣
 ↪for w in x if not w in set_of_stopwords])
```

[235]:
```
num_removed = len(dftrain['review_words'][0]) -␣
 ↪len(dftrain['review_meaningful_words'][0])
print('For the first review entry, the number of stop words removed is {0}.'.
 ↪format(num_removed))
```

```
For the first review entry, the number of stop words removed is 218.
```

We can also stem the words with PorterStemmer() and Lemmatizer() to consider only word stems. But for this training data, better results was produced without stemming.

[173]:
```
from nltk.stem import PorterStemmer, WordNetLemmatizer
```

[175]:
```
porter_stemmer = PorterStemmer()
wordnet_lemmatizer = WordNetLemmatizer()
```

[244]:
```
num_removed = len(dftrain['review_words'][0]) -␣
  ↪len(dftrain['review_meaningful_words'][0])
print('For the first review entry, the number of stop words removed is {0}.'.
  ↪format(num_removed))
```

```
For the first review entry, the number of stop words removed is 218.
```

[248]:
```
dftrain['review_cleaned'] = dftrain['review_meaningful_words'].apply(lambda x:␣
  ↪' '.join(x)) # comment if using stemming
```

[250]:
```
#Add revies_cleaned as a new column to the training data.
dftrain.drop(['review', 'review_bs', 'review_letters_only', 'review_words',␣
  ↪'review_meaningful_words'],
             axis=1, inplace=True)
display(dftrain.head())
```

```
         id  sentiment predicted_sentiment vader_sentiment  \
0  "5814_8"          1            Positive        negative
1  "2381_9"          1            Positive        positive
2  "7759_3"          0            Negative        negative
3  "3630_4"          0            Positive        negative
4  "9495_8"          1            Negative        positive

                                      review_cleaned
0  stuff going moment mj started listening music …
1  classic war worlds timothy hines entertaining …
2  film starts manager nicholas bell giving welco…
3  must assumed praised film greatest filmed oper…
4  superbly trashy wondrously unpretentious explo…
```

[252]:
```
print(dftrain['review_cleaned'][0])
```

```
stuff going moment mj started listening music watching odd documentary watched
wiz watched moonwalker maybe want get certain insight guy thought really cool
eighties maybe make mind whether guilty innocent moonwalker part biography part
feature film remember going see cinema originally released subtle messages mj
feeling towards press also obvious message drugs bad kay visually impressive
course michael jackson unless remotely like mj anyway going hate find boring may
call mj egotist consenting making movie mj fans would say made fans true really
nice actual feature film bit finally starts minutes excluding smooth criminal
sequence joe pesci convincing psychopathic powerful drug lord wants mj dead bad
beyond mj overheard plans nah joe pesci character ranted wanted people know
```

supplying drugs etc dunno maybe hates mj music lots cool things like mj turning car robot whole speed demon sequence also director must patience saint came filming kiddy bad sequence usually directors hate working one kid let alone whole bunch performing complex dance scene bottom line movie people like mj one level another think people stay away try give wholesome message ironically mj bestest buddy movie girl michael jackson truly one talented people ever grace planet guilty well attention gave subject hmmm well know people different behind closed doors know fact either extremely nice stupid guy one sickest liars hope latter

0.1 Creating Features from a Bag of Words (Using scikit-learn)

[255]:
```
#Initialize the CountVectorizer object, which is scikit-learn's bag of words␣
↪tool. CountVectorizer converts a collection of text documents to a matrix of␣
↪token counts.
vectorizer = CountVectorizer(analyzer="word", preprocessor=None,␣
↪tokenizer=None, stop_words=None, max_features=5000)
```

analyzer="word" indicates the feature we are using are words; preprocessor=None, tokenizer=None and stop_words=None mean the data needes no more preprocessing, tokenization and removing stop sords since we've already done these in the Data Cleaning and Text Processing step; max_features=5000 means we only take the top 5000 frequent words as our words in the bag thus limiting the size of the feature vector and speeding up the modeling process. fit_transform() method does two functions: First, it fits the model and learns the vocabulary; second, it transforms our training data into feature vectors. The input to fit_transform should be a list of strings.

[260]:
```
train_data_features = vectorizer.fit_transform(list(dftrain['review_cleaned'].
↪values))

# Numpy arrays are easy to work with, so convert the result to an array
train_data_features = train_data_features.toarray()
```

The resulting train_data_features is a array which contains the occurrence of bag of words in our training data. Take a look at the first row.

[263]:
```
train_data_features[0]
```

[263]:
```
array([0, 0, 0, …, 0, 0, 0])
```

[265]:
```
print('The dimension of train_data_features is {}.'.format(train_data_features.
↪shape))
```

```
The dimension of train_data_features is (25000, 5000).
```

0.2 Modularizing the Cleaning Process

[270]:
```
def clean_reviews(reviews, remove_stopwords=False, stem=False):
    """
    to clean review strings
    review: a list of review strings
    remove_stop_words: whether to remove stop words
    output: a list of clean reviews
    """
    # 1. Remove HTML
    reviews_text = list(map(lambda x: BeautifulSoup(x, 'html.parser').
 ↪get_text(), reviews))
    #
    # 2. Remove non-letters
    reviews_text = list(map(lambda x: re.sub("[^a-zA-Z]"," ", x), reviews_text))
    #
    # 3. Convert words to lower case and split them
    words = list(map(lambda x: x.lower().split(), reviews_text))
    #
    # 4. Optionally remove stop words (false by default)
    if remove_stopwords:
        set_of_stopwords = set(stopwords.words("english"))
        meaningful_words = list(map(lambda x: [w for w in x if not w in␣
 ↪set_of_stopwords], words))

    # 5. Optionally stem the words
    if stem:
        porter_stemmer = PorterStemmer()
        wordnet_lemmatizer = WordNetLemmatizer()
        stemmed_words = list(map(lambda x: [porter_stemmer.stem(w) for w in x],␣
 ↪meaningful_words))
        stemmed_words = list(map(lambda x:[wordnet_lemmatizer.lemmatize(w) for␣
 ↪w in x], stemmed_words))

        # 6. Join the words to a single string
        clean_review = map(lambda x: ' '.join(x), stemmed_words)
    else:
        clean_review = list(map(lambda x: ' '.join(x), meaningful_words))

    return clean_review
```

0.3 Load and Clean Test Data

[273]:
```
# Read the test data
test = pd.read_csv('testData.tsv', header=0, delimiter='\t', quoting=3)

# Verify that there are 25,000 rows and 2 columns
```

```python
print('The dimension of test data is {}.'.format(test.shape))

# Get a bag of words for the test set, and convert to a numpy array
clean_test_reviews = clean_reviews(list(test['review'].values),␣
 ↪remove_stopwords=True)
test_data_features = vectorizer.transform(clean_test_reviews)
test_data_features = test_data_features.toarray()
```

```
The dimension of test data is (25000, 2).
```

```
/var/folders/18/ygzmjvj90m72g8_1d1sfk1b80000gn/T/ipykernel_1364/2521139183.py:9:
MarkupResemblesLocatorWarning: The input looks more like a filename than markup.
You may want to open this file and pass the filehandle into Beautiful Soup.
  reviews_text = list(map(lambda x: BeautifulSoup(x, 'html.parser').get_text(),
reviews))
```